

DINESH PHUYAL

2181 McCarty Hall A PO Box 110290 Gainesville, FL 32611

Email: phuyaldinesh@ufl.edu Alternative Email: phuyaldinesh@gmail.com

CURRENT POSITION (05/2025– To present)

Postdoctoral associate, Department of Soil, Water, and Ecosystem Sciences | University of Florida | College of Agricultural and Life Sciences | United States.

Research: Soil Nutrient Management (Focusing on phosphorus). [Link](#)

PREVIOUS UNIVERSITY EDUCATION

1. **Doctor of Philosophy, Agronomy | Soil and Crop Sciences Department | Texas A&M University | College of Agriculture and Life Sciences | United States.** GPA: 4.00 [Link](#)
Ph.D. Dissertation Title: Investigating the Biological Nitrification Inhibition Activity in Selected Sorghum Genotypes.

Duties: Graduate Research Assistant (05/2020- 05/2025)

- Focus: Biological Nitrification Inhibition (BNI) in Sorghum.
- Designed experiments, collected, and analyzed data to investigate nitrification rates, nitrous oxide emissions, and nitrogen use efficiency in field and greenhouse conditions.
- Utilized LICOR gas analyzers, Spectrophotometers, Elementar, DNA extraction and purification kits for soil nutrient, soil microbial, greenhouse gas, and plant nutrient analysis.

Supervisory Committee:

Dr. Nithya Rajan and Dr. Sakiko Okumoto (Chair)

Dr. William L. Rooney

Dr. Sanjay Antony-Babu

2. **Master of Science, Horticulture | University of Florida | College of Agricultural and Life Sciences | United States.** GPA: 4.00 [Link](#)
M.Sc. Thesis Title: Unraveling the effect of advanced horticultural practices on Huanglongbing-affected grapefruit.

Duties: Graduate Research Assistant (01/2018-05/2020)

- Focus: Soil Nutrient Management and Advanced Horticultural Practices on Huanglongbing-affected Grapefruit.
- Managed field trials and performed data collection and analysis to investigate the effects of planting density and enhanced nutrition on tree health, fruit yield, and quality.
- Operated field instruments as well as data collection tools such as soil moisture sensors and the LICOR 6400.

Supervisory committee:

Dr. Rhuanito Soranz Ferrarezi (Chair)

Dr. Davie M. Kadyampakeni

Dr. Kelly Morgan

3. **Bachelor of Science | Agriculture | Tribhuvan University | Institute of Agriculture and Animal Sciences | Nepal.** Grade: Distinction [Link](#)

Relevant coursework: Soil science, Agronomy, Horticulture, Statistics, Plant Physiology, Genetics and Plant Breeding, Biotechnology, Entomology, Plant Pathology, Biochemistry, Microbiology, Fodder production, Agricultural Economics, Social Mobilization, Extension education.

PEER-REVIEWED JOURNAL ARTICLES (PUBLISHED)

1. Nair, V. D., **Phuyal, D.**, & Vardanyan, L. (2026). Bioavailable phosphorus across Florida's diverse soil orders: implications for crop productivity and environmental protection [Original Research]. *Frontiers in Environmental Science*, Volume 14 - 2026.
<https://doi.org/10.3389/fenvs.2026.1748291>
2. Watson, K. L., Gignac, J. E., Navarrete, A. A., Joharzadeh, L., Osei, B., **Phuyal, D.**, . . . Wyatt, B. M. (2026). Low-intensity prescribed fire has no immediate impact on soil physical and chemical properties or enzymatic activity in the Post Oak Savanna ecoregion, Texas, USA. *Agrosystems, Geosciences & Environment*, 9(1), e70315.
<https://doi.org/10.1002/agg2.70315>
3. Sapkota, D., & **Phuyal, D.** (2026). Agricultural Runoff and Waterborne Disease in Primary Care: A Review. *International Journal of Environmental Medicine*, 1(1), 5.
<https://www.mdpi.com/3042-7681/1/1/5>
4. **Phuyal, D.**, & Nair, V. D. (2025). The dynamics, analysis, and sustainable management of phosphorus in muck soils: A review. *Frontiers in Environmental Science*, 13, 1703620.
<https://doi.org/10.3389/fenvs.2025.1703620>
5. Ghosh, E., Rajan, N., **Phuyal, D.**, Subramanian, N., & Bagavathiannan, M. (2024). High rhizospheric ammonium levels in Sorghum halepense (johnsongrass) suggest nitrification inhibition potential. *Agricultural & Environmental Letters*, 9(2), e20137.
<https://doi.org/10.1002/acl2.20137>
6. **Phuyal, D.**, Nogueira, T. A. R., Jani, A. D., Kadyampakeni, D. M., Morgan, K. T., & Ferrarezi, R. S. (2020). 'Ray Ruby' grapefruit affected by Huanglongbing I: Planting density and soil nutrient management. *HortScience*, 55(9), 1411–1419.
<https://doi.org/10.21273/HORTSCI15111-20>
7. **Phuyal, D.**, Nogueira, T. A. R., Jani, A. D., Kadyampakeni, D. M., Morgan, K. T., & Ferrarezi, R. S. (2020). 'Ray Ruby' grapefruit affected by Huanglongbing II: Planting density, soil, and foliar nutrient management. *HortScience*, 55(9), 1420–1432.
<https://doi.org/10.21273/HORTSCI15255-20>

PEER-REVIEWED JOURNALS (SUBMITTED)

1. Mechanistic insight into biochar effects on soil physical, chemical, and biological properties: A Review. Arjun Kafle, Kaushik Adhikari, M. Samrat Dahal, Anil Timilsina, **Dinesh Phuyal** [Submitted: Discover Soil]
2. **Dinesh Phuyal**, Musfiq SK Us-Salehin, Sanjaya Antony-Babu, Sakiko Okumoto, William Rooney, & Nithya Rajan. Nitrous oxide emission from croplands: Biogeochemical drivers, nitrification inhibitors, policy implications, and future directions[Submitted: Reviews in Environmental Science and Bio/Technology]
3. Challenges in using Mehlich 3 as Florida's soil test for phosphorus Vimala D. Nair, **Dinesh Phuyal**, & Lilit Vardanyan [Submitted: EDIS]

PEER-REVIEWED JOURNALS (IN PREPARATION)

1. Fertilizer Types Impacts Biological Nitrification Inhibition of Sorghum and its Climate-smart Benefits. [In preparation- PI is reviewing]
2. Unraveling the Effect of Biological Nitrification Inhibition of Sorghum Under Different Soil Types. [In preparation- PI is reviewing]
3. The Biological Nitrification Inhibition Activity of Sorghum Shows a Potential Climate-Smart Solution for Sustainable Agriculture in a Field Condition. [In preparation- PI is reviewing]

EXTENSION AND OUTREACH PUBLICATION

1. **Phuyal, D.** (2026). Why too much phosphorus in America's farmland is polluting the country's water. <https://doi.org/10.64628/AAI.yrhd77xxd>
2. **Phuyal, D.**, & Davenport, R. (2022). Science Advocacy and Leadership Opportunities for Graduate Students. *CSA News*, September.
[DOI: 10.1002/csan.20867](https://doi.org/10.1002/csan.20867)
3. **Phuyal, D.**, & Ferrarezi, R. (2020). Improved Horticultural Practices for Grapefruit Production in the Indian River District. *Indian River Citrus League, Newsletter*, September.
[Link](#)

4. Ferrarezi, R., **Phuyal, D.**, Kadyampakeni, D., & Morgan, K. (2020). Effect of Planting Density and Enhanced Nutrition on Grapefruit. EDIS, January.
[Link](#)
5. Biological nitrification inhibition: A climate-smart solution and key for sustainability. [In preparation- PI is reviewing]

JOURNALS REVIEWED

1. Frontiers in Plant Science (1 article)
2. Computers and Electronics in Agriculture (1 article)
3. Hort Science (1 article)
4. Agrosystems, Geosciences & Environment (3 articles)
5. Agronomy (7 articles)
6. Biology (3 articles)
7. Forests (8 articles)
8. Nitrogen (3 article)

CONFERENCE ABSTRACTS (* is the presenting author)

1. **Phuyal, D.***, Nair, V. D., Sadeghibaniani S., Vardanyan, L., Sandoya, G., Hailegnaw, N., & Birhan D. Refining Phosphorus Diagnostics in Organic Soils: Bridging Agronomic Needs and Water Quality Goals. 10th biennial UF Water Institute Symposium. Gainesville, Florida. February 24-26. (*Poster Presentation*)
[Link](#)
2. Portmess, A.*, Ballou, J., **Phuyal, D.**, Vardanyan, L., Sandoya, G., Hailegnaw, N., & Nair, V. D. Water Quality Impacts of Phosphorus from Soil Amendments in Florida's Sandy Soils. 10th biennial UF Water Institute Symposium. Gainesville, Florida. February 24-26. (*Poster Presentation*)
[Link](#)
3. **Phuyal, D.***, Nair, V. D., Kadyampakeni, D. M., Atta, A. A., & Vardanyan, L. (2025). Soil Phosphorus Storage Capacity and Bioavailable Phosphorus Indicators as Predictors of Citrus Yield in Florida Soils. ASA, CSSA, SSSA International Annual Meeting, Salt Lake City, UT, November 9-12. (*Poster Presentation*)
[Link](#)
4. Nair, D. V.*, Vardanyan, L., & **Phuyal, D.** (2025). Optimizing Phosphorus Management: Regional Assessment of Extraction Methods in Florida's Acid-Mineral Soils. ASA, CSSA, SSSA International Annual Meeting, Salt Lake City, UT, November 9-12. (*Oral Presentation*)
[Link](#)
5. Okumoto, S.*, Maharjan, B. K., Jeong, U., Leon, F., Rooney, W., Baerson, S., Rajan, N., & **Phuyal, D.** (2025). Enhancing Biological Nitrification Inhibition Activities in Sorghum. ASA, CSSA, SSSA International Annual Meeting, Salt Lake City, UT, November 9-12. (*Oral Presentation*)
[Link](#)
6. Gomez, C.*, Miyanaka, N., Salehin, S. M. U., **Phuyal, D.**, Antony-Babu, S., Okumoto, S., Rooney, W. L., & Rajan, N. (2025). Reducing Nitrogen Loss in Sorghum Cropping Systems through Biological Nitrification Inhibition (BNI). ASA, CSSA, SSSA International Annual Meeting, Salt Lake City, UT, November 9-12. (*Oral Presentation; presenter won first prize*)
[Link](#)
7. Gomez, C.*, Yuan, P., Miyanaka, N., Salehin, S. M. U., **Phuyal, D.**, Rooney, W. L., Okumoto, S., Antony-Babu, S., & Rajan, N. (2025). Reducing Nitrate Leaching through Nitrogen Retention Using Biological Nitrification Inhibition (BNI) in Sorghum. ASA, CSSA, SSSA International Annual Meeting, Salt Lake City, UT, November 9-12. (*Poster Presentation; presenter won first prize*)
[Link](#)
8. **Phuyal, D.***, Rajan, N., Rooney, W. L., Antony-Babu, S., Okumoto, S., Peiguo, Y., Casey, K. D., & Subbarao, G. V. (2024). Evidence for Biological Nitrification Inhibition Activity of Sorghum (*Sorghum bicolor*) at Field: Implications for Climate Change and Agriculture. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX, November 10-13. (*Oral Presentation*)
[Link](#)

9. **Phuyal, D.***, Rajan, N., Rooney, W. L., Antony-Babu, S., Okumoto, S., Peiguo, Y., Casey, K. D., & Subbarao, G. V. (2024). Genotype and Soil Interactions on Biological Nitrification Inhibition Activity of Sorghum (*Sorghum bicolor*). ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX, November 10-13. (Oral Presentation)
[Link](#)
10. Gomez, C.*, **Phuyal, D.**, Miyataka, N., Salehin, S. M. U., Rooney, W. L., Okumoto, S., & Rajan, N. (2024) The Effect of Fertilizer Type and Genotype on Biological Nitrification Inhibition Expression of Sorghum Under Field Conditions. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX, November 10-13. (Poster Presentation)
[Link](#)
11. **Phuyal, D.***, Rajan, N., Rooney, W. L., Antony-Babu, S., Casey, K. D., Okumoto, S., & Subbarao, G. V. (2023). Exploring the Impact of Sorghum BNI Activity: Implications for Climate Change and Agriculture. ASA, CSSA, SSSA International Annual Meeting, St Louis, MO, 29 October to 1 November. (Oral Presentation; presenter won first prize)
[Link](#)
12. **Phuyal, D.***, Rajan, N., Rooney, W. L., Antony-Babu, S., Casey, K. D., Okumoto, S., & Subbarao, G. V. (2023). Impact of Nitrogen Fertilizer Sources on Biological Nitrification Inhibition Activity of Sorghum. ASA, CSSA, SSSA International Annual Meeting, St Louis, MO, 29 October to 1 November. (Oral Presentation)
[Link](#)
13. Schill, M. L.*, Subramanian, N. K., **Phuyal, D.**, Rajan, N., & Bagavathiannan, M. (2023). *Sorghum halepense* Promotes Biological Nitrification Inhibition in Its Rhizosphere. ASA, CSSA, SSSA International Annual Meeting, St Louis, MO, 29 October to 1 November. (Poster Presentation)
[Link](#)
14. Schill, M. L.*, **Phuyal, D.**, Chobhe, K., Rajan, N., Ibrahim, A., & Bagavathiannan, M. (2023). Investigating the Biological Nitrification Inhibition Potential in *Avena sativa* Cultivars. ASA, CSSA, SSSA International Annual Meeting, St Louis, MO, 29 October to 1 November. (Oral Presentation; presenter won third prize)
[Link](#)
15. **Phuyal, D.***, Rajan, N., Rooney, W. L., Antony-Babu, S., Casey, K. D., Zapata, D., Schill, M., Subramanian, N. K., Salehin, S. M. U., Maharjan, B. K., Okumoto, S., & Subbarao, G. V. (2022). Biological Nitrification Inhibition Activity of Sorghum Genotypes with Differential Sorgoleone Secretion Capacity. ASA, CSSA, SSSA International Annual Meeting, Baltimore, MD, 6-9 November. (Oral Presentation)
[Link](#)
16. **Phuyal, D.***, Rajan, N., Rooney, W. L., Antony-Babu, S., Schill, M., Subramanian, N. K., Okumoto, S., & Subbarao, G. V. (2022). Effect of Biological Nitrification Inhibition Activity of Sorghum Genotypes on Nitrification in a Field Study. ASA, CSSA, SSSA International Annual Meeting, Baltimore, MD, 6-9 November. (Poster Presentation; presenter won second prize)
[Link](#)
17. Schill, M. L.*, **Phuyal, D.**, Maharjan, B. K., Rajan, N., Subramanian, N. K., & Bagavathiannan, M. (2022). Biological Nitrification Inhibition Potential Improves Nitrogen Availability in Johnsongrass (*Sorghum halepense*) Rhizosphere. ASA, CSSA, SSSA International Annual Meeting, Baltimore, MD, 6-9 November. (Oral Presentation)
[Link](#)
18. **Phuyal, D.***, Rajan, N., Rooney, W. (2022). Sorghum Genotypes with Contrasting Sorgoleone Secretion Convey Varying Biological Nitrification Inhibition (BNI) Activity in Acidic Soil. 8th Annual Texas A&M Plant Breeding Symposium, Soil and Crop Sciences, Texas A & M University, College Station, TX, 17 February. (Poster Presentation)
19. Ghosh, E.*, Rajan, N., **Phuyal, D.**, Bagavathiannan, M., Subramanian, N. (2022). Preliminary Investigation of Biological Nitrification Inhibition (BNI) Potential in Johnsongrass (*Sorghum halepense*). 62nd Meeting of Weed Science Society of America, Virtual Meeting, 22-24 February. (Poster Presentation; presenter won first prize)
[Link](#)
20. **Phuyal, D.***, Rajan, N., Schnell, R., Rooney, W., Maharjan, B., Casey, K., Zapata, D., Okumoto, S., Subramanian, N., Kim, J., Chu, K. H., Peterson, J. A., & Subbarao, G. V. (2021). Biological Nitrification Inhibition (BNI) Activity of Sorghum Genotypes on a Typical Field Soil. ASA, CSSA, SSSA International Annual Meetings, Salt Lake City, UT, 7-10

November. (*Oral Presentation*)

[Link](#)

21. **Phuyal, D.***, Rajan, N., Schnell, R., Salehin, S. M., Casey, K., Rooney, W., Maharjan, B., Okumoto, S., & Subbarao, G. V. (2021). Effect of Biological Nitrification Inhibition of Sorghum on Nitrous Oxide Emission Under Different Nitrogen Fertilizer Types. ASA, CSSA, SSSA International Annual Meetings, Salt Lake City, UT, 7-10 November. (*Poster Presentation*)
[Link](#)
22. Rajan, N.*, **Phuyal, D.**, Maharjan, B., Okumoto, S., Casey, K., Subramanian, N., Peterson, J. A., Bagavathiannan, M., Jifon, J., Rooney, W., Schnell, R., & Subbarao, G. V. (2020). Climate Solutions: Biological Nitrification Inhibition in Modern Crop Varieties. 2020. AGU Virtual Meeting, 15 December 2020. (*Oral Presentation*)
[Link](#)
23. **Phuyal, D.***, Rajan, N., Rooney, W., Kim, J., Chu, K. H., Subramanian, N., Maharjan, B., Okumoto, S., Schnell, R., Peterson, J. A., & Subbarao, G. V. (2020). Effect of Biological Nitrification Inhibition (BNI) of Sorghum on Weswood Silt Loam Soil. 2020 Texas Plant Protection Conference. ASA, CSSA, SSSA International Annual Meetings, Virtual Meeting, 8 December. (*Poster Presentation*)
[Link](#)
24. **Phuyal, D.***, Rajan, N., Rooney, W., Kim, J., Chu, K. H., Subramanian, N., Maharjan, B., Okumoto, S., Schnell, R., Peterson, J. A., & Subbarao, G. V. (2020). Climate Smart Farming: A Preliminary Investigation of Biological Nitrification Inhibition (BNI) in Selected Sorghum Genotypes. ASA, CSSA, SSSA International Annual Meetings, Virtual Meeting, 9-13 November. (*Poster Presentation; presenter won second prize*)
[Link](#)
25. **Phuyal, D.***, Nogueira, T. A. R., Jani, A. D., Kadyampakeni, D., Morgan, K., & Ferrarezi, R. S. (2020). Tree Density and Soil Micronutrient Application Using Controlled-release Fertilizer on Grapefruit Affected by Huanglongbing. 117th Annual American Society for Horticultural Science, Orlando, FL, 9-13 August. (*Poster Presentation*)
[Link](#)
26. **Phuyal, D.***, Nogueira, T. A. R., Jani, A. D., Kadyampakeni, D., Morgan, K., & Ferrarezi, R. S. (2020). Effect of Tree Planting Density and Nutrient Application by Soil and Foliar on Grapefruit Trees Affected by Huanglongbing. 117th Annual American Society for Horticultural Science, Orlando, FL, 9-13 August. (*Oral Presentation*)
[Link](#)
27. Rajan, N.*, Maharjan, B., Okumoto, S., **Phuyal, D.**, Casey, K., Subramanian, N., Peterson, J. A., Bagavathiannan, M., Jifon, J., Rooney, W., Schnell, R., & Subbarao, G. V. (2020). Biological Nitrification Inhibition in Sorghum: A Forgotten Trait with Potential for Greenhouse Gas Mitigation. 5th Annual Meeting of the Soil Health Institute. 30 July 2020. (*Oral Presentation*)
[Link](#)
28. Ferrarezi, R. S.*, **Phuyal, D.**, Kadyampakeni, D., & Morgan, K. T. (2019). Can Higher Planting Density and Enhanced Fertilization Increase Fruit Yield of HLB-infected Grapefruit? Materials Innovation for Sustainable Agriculture Center of Excellence, Orlando, FL, 24 October. (*Oral Presentation*)
[Link](#)
29. Ferrarezi, R. S.*, **Phuyal, D.**, Kadyampakeni, D., & Morgan, K. T. (2019). Grapefruit Nutritional Research in the Indian River District. IRREC Citrus Nutrition Management Day, Fort Pierce, FL, 20 October. (*Oral Presentation*)
30. **Phuyal, D.***, Ferrarezi, R. S., Morgan, K. T., & Kadyampakeni, D. (2019). Tree Density and Micronutrient Application on Grapefruit Affected by Huanglongbing. 116th Annual American Society for Horticultural Science, Las Vegas, NV, 21-25 July. (*Oral Presentation*)
[Link](#)
31. **Phuyal, D.***, Ferrarezi, R. S., Morgan, K. T., & Kadyampakeni, D. M. (2019). Assessment of Different Grapefruit Tree Spacing on Tree Health, Fruit Quality, and Fruit Yield. 7th Annual South Florida Graduate Research Symposium, Homestead, FL, 16 July. (*Poster Presentation*)
32. **Phuyal, D.***, Ferrarezi, R. S., Morgan, K. T., & Kadyampakeni, D. (2019). Nutrient Management on HLB-affected Grapefruit on Flatwoods Soil in the Indian River District. Florida

State Horticultural Society 132nd Annual Conference, Orlando, FL, 9-11 June. (*Oral Presentation*)

INVITED PRESENTATION

Phuyal, D., and Rajan, N. (2022). Update on Sorghum BNI Research at Texas A&M. The 4th Biennial Meeting organized by Japan International Research Center for Agricultural Sciences, Tsukuba, JAPAN, 17 November to 19 November. (Oral Presentation)

[Link](#)

SPECIAL INVITATION

2025 Student Commencement Speaker, Texas A&M University, College of Agriculture and Life Sciences: Selected to deliver the Ph.D. appreciation speech at the graduation ceremony. [Link](#)

PROFESSIONAL MEMBERSHIP AND LEADERSHIP

2022- 2023	Vice President: Soil and Crop Sciences Graduate Organization
2021-2022	International Student Association (External Programs Committee)
2020-2021	NAPA Career and Outreach Committee
2019-Present	Crop Science Society of America
2019-Present	American Society of Agronomy
2018-2020	UF Treasure Coast Graduate Student Organization
2018-2020	Florida State Horticultural Society
2018-2020	American Society for Horticultural Science

FELLOWSHIP AND AWARDS

2022-23	<i>Encompass Scholar (ASA-CSSA-SSSA)</i> : Participated in a professional development workshop.
2022	<i>ASA-CSSA-SSSA SEED Ambassador (Scientists Engaging & Educating Decision-makers Ambassador)</i> : Participated in an advocacy training program to connect scientists with members of Congress.
2022-23	<i>Future Leaders in Science</i> : Selected to participate in the 2022 Virtual Congressional Visits Day. Visited the offices of Sen. Ted Cruz, Rep. Jake Ellzey, Rep. Roger Williams, and Rep. Henry Cuellar to discuss USDA funding for agricultural science.
2019	<i>Yara North America Crop Innovation Scholarship</i> : University of Florida College of Agricultural and Life Sciences: Awarded for outstanding academic and research record.
2019	<i>Center for Stress Resilient Agriculture, University of Florida</i> : To participate in an innovative root course.
2018	<i>LI-COR Biosciences Training Travel Award</i> : Participated in LICOR 6400 photosynthesis equipment training.
2012-2016	<i>Tribhuvan University Scholarship</i> : Awarded for outstanding academic performance across all eight semesters.
2012	<i>Tribhuvan University- TU Merit Admission Scholarship</i> : Awarded for securing a high score in the admission exam.

TEACHING EXPERIENCE

- Teaching Assistant, Soil Fertility and Plant Nutrient Management Laboratory (SCSC 432), Texas A & M University, United States**
Fall 2024: Assist with the laboratory component for fourteen undergraduate students, focusing on methods used in soil testing, fertilizer recommendations, and the chemical and physical properties of soils. Guide the determination and analysis of specific soil characteristics and support students in understanding and applying these techniques.
- Teaching Assistant, Soil Science (SCSC 301), Texas A & M University, United States**
Spring 2022: Supported both virtual and in-person instruction for fifty undergraduate students. Facilitated laboratory sessions that involved soil physical, chemical, and biological analysis and guided students in the interpretation of results. Assisted with the scientific evaluation of soil properties and their environmental roles. Also, proctored exams.
- Teaching Assistant, World Food and Fiber Crops (SCSC 105), Texas A & M University, United States**

Fall 2021: Assisted twenty-four undergraduate students in both lecture and laboratory settings. The course covered plant relationships, structure, development, and the environmental factors affecting crops. Supported students in plant data collection, statistical analysis, and report presentation, focusing on the technological aspects of agricultural practices and food production.

4. **Agriculture Instructor, Galkot Higher Secondary School, Baglung, Nepal**
2016 - 2017: Led an introductory entomology and vegetable production course for 30 high school students, fostering foundational knowledge in agricultural sciences.

EXTENSION ACTIVITIES

1. **10th biennial UF Water Institute Symposium Meeting: Poster presentation Judge**
February 2026: Served as a Judge for six graduate students poster competition related to water quality.
2. **CANVAS Meeting: Oral presentation Judge**
November 2025: Served as a Judge for eight graduate oral competitions on the Soil Carbon & Greenhouse Gas Emissions Community.
3. **CANVAS Meeting: Poster presentation Judge**
November 2025: Served as a Judge for five graduate poster competitions on the ASA Nutrients and Environmental Quality Community
4. **ASA, CSSA, SSSA International Annual Meeting: Poster Judge**
November 2024: Served as a Judge for five undergraduate poster competitions on Soil Nutrition and Plant Physiology.
5. **Student Research Week, Texas A&M University**
March 2023: Served as a Judge, grading and scoring six posters from undergraduate and graduate students.
6. **Student Research Week, Texas A&M University**
January 2022: Served as a Judge, grading and scoring eight posters from undergraduate students.
7. **Florida Citrus Show**
January 2018 and 2019: Assisted at the registration desk and helped presenters load presentations. Ensured the audio and visual equipment functioned properly for over thirty presenters, serving over eight hundred attendees.
8. **Agricultural Workers Training**
January 2018: Interacted with growers and assisted them during the event. Set up and took down materials for the seminar, serving over three hundred attendees.
9. **Citrus Expo**
August 2019: Prepared and presented two posters at a grower's event:
 - Tree Density and Micronutrient Application on Grapefruit Affected by Huanglongbing.
 - Application of Soil Moisture Sensors.

ADVOCACY ACTIVITIES

Congressional Meetings 2022

- Selected among 14 SEED Ambassadors nationwide by the American Society of Agronomy.
- Trained for a year to build trusted relationships between Science Societies and the Members of Congress.
- Communicated with some Congressional Members (offices of Sen. Ted Cruz, Rep. Jake Ellzey, Rep. Roger Williams, and Rep. Henry Cuellar) via emails/phone and video calls and requested to support agricultural research and funding.
- Discussed the impact of USDA-funded research programs in addressing current and emerging agricultural problems.

CERTIFICATION

- FAA Part 107 UAV pilot License (Valid until Jan 2027) [Link](#)
- An Introduction to Evidence-Based Undergraduate STEM Teaching (From 2025 to Unexpired) [Link](#)

TRAVEL GRANTS

2022	<i>The Lloyd and Maxine Rooney Fellowship Endowment</i> : Presented research at an international conference.
2021	<i>The Lloyd and Maxine Rooney Fellowship Endowment</i> : International Travel Scholarship (could not be used due to the postponement of the conference).
2021	<i>Graduate Student Leadership Conference</i> : Participated in leadership training.
2019	<i>American Society for Horticultural Science</i> : Presented research at the ASHS international conference.
2019	<i>Graduate Student Council Travel Award</i> : Presented research at the ASHS international conference.